Chapter 1: The Real and Complex Number Systems

Section: The Real Field

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

The Existence Theorem

Archimedean Property and Density of $\mathbb{Q}$

Existence of nth Roots

Corollary and Decimal Expansions

Summary

The Existence Theorem

Why This Section Matters

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- The rationals $\mathbb{Q}$ have gaps (e.g., no $p \in \mathbb{Q}$ with $p^2 = 2$)
- We need an ordered field with the least-upper-bound property
- This section: state the existence of $\mathbb{R}$ and derive key consequences

Theorem 1.19: Existence of $\mathbb{R}$

Theorem 1.19

There exists an ordered field $\mathbb{R}$ which has the least-upper-bound property.

Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.

Theorem 1.19: What This Means

The members of $\mathbb{R}$ are called real numbers.

Key points:

- $\mathbb{Q} \subset \mathbb{R}$, with the same $+$ and $\times$ on rationals

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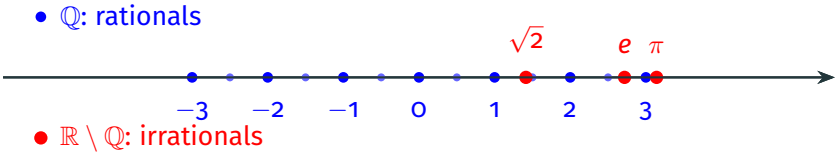
Key points:

- $\mathbb{Q} \subset \mathbb{R}$, with the same $+$ and $\times$ on rationals
- Positive rationals remain positive in $\mathbb{R}$
- The proof *constructs* $\mathbb{R}$ from $\mathbb{Q}$ (via Dedekind cuts — see Appendix)

Theorem 1.19: Visualizing $\mathbb{R}$ and $\mathbb{Q}$



$\mathbb{R}$ fills all the gaps in $\mathbb{Q}$ — it is a *complete* ordered field.

Archimedean Property and Density of $\mathbb{Q}$

Theorem 1.20: Two Key Consequences

Theorem 1.20(a) — Archimedean Property

If $x \in \mathbb{R}$, $y \in \mathbb{R}$, and $x > 0$, then there is a positive integer n such that

$$nx > y.$$

Theorem 1.20: Two Key Consequences

Theorem 1.20(a) — Archimedean Property

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Theorem 1.20(b) — Density of $\mathbb{Q}$ in $\mathbb{R}$

If $x \in \mathbb{R}$, $y \in \mathbb{R}$, and $x < y$, then there exists $p \in \mathbb{Q}$ such that

$$x < p < y.$$

Terminology

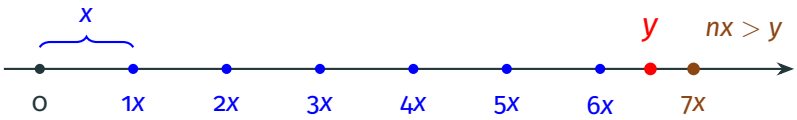
- Part (a) is the archimedean property of $\mathbb{R}$:

  No element of $\mathbb{R}$ is "infinitely large" relative to any other.

- Part (b) says $\mathbb{Q}$ is dense in $\mathbb{R}$:

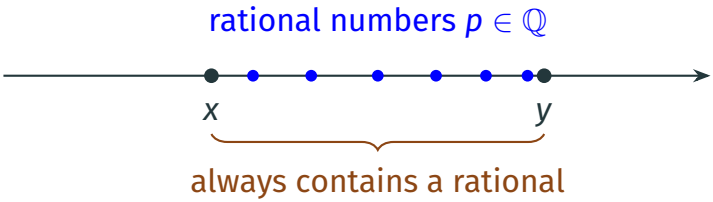
  Between any two reals, there is a rational.

Archimedean Property: Illustration



No matter how small $x > 0$ is, repeated addition eventually surpasses any y.

Density of $\mathbb{Q}$ in $\mathbb{R}$: Illustration



Proof of 1.20(a): Strategy

Goal: Show that for $x > 0$, the set $\{nx : n \in \mathbb{Z}^+\}$ is unbounded above.

Technique: Proof by contradiction using the least-upper-bound property.

Idea:

1. Suppose the set $A = \{nx\}$ is bounded above.
2. Then $\alpha = \sup A$ exists (by Theorem 1.19).
3. Show this leads to a contradiction.

Proof of 1.20(a)

Step 1: Setup

Let $A = \{nx : n \in \mathbb{Z}^+\}$.

Suppose for contradiction that A is bounded above (i.e., $nx \leq y$ for all n).

By Theorem 1.19, $\alpha = \sup A$ exists.

Proof of 1.20(a)

Step 1: Setup

Let $A = \{nx : n \in \mathbb{Z}^+\}$. Suppose $\alpha = \sup A$ exists.

Step 2: The contradiction

Since $x > 0$, we have $\alpha - x < \alpha$.

So $\alpha - x$ is *not* an upper bound of A.

Hence $\alpha - x < mx$ for some positive integer m.

But then $\alpha < (m+1)x \in A$. Contradiction!

Proof of 1.20(b): Strategy

Goal: Given $x < y$, find a rational $p = m/n$ with $x < m/n < y$.

Idea: Use the archimedean property *twice*:

1. Find n large enough so that $1/n$ is smaller than $y - x$.
2. Find the right integer m so that m/n lands between x and y.

Proof of 1.20(b)

Step 1: Make the mesh fine enough

Since $y - x > 0$, by part (a) there exists a positive integer n with

$$n(y - x) > 1.$$

Proof of 1.20(b)

Step 1: Make the mesh fine enough

Find $n \in \mathbb{Z}^+$ with $n(y - x) > 1$.

Step 2: Bound nx between integers

By part (a), there exist positive integers m_1, m_2 with:

$$m_1 > nx \quad \text{and} \quad m_2 > -nx.$$

Hence $-m_2 < nx < m_1$, so there is an integer m with

$$m - 1 \leq nx < m.$$

Proof of 1.20(b)

Step 3: Combine the inequalities

From $nx < m$ and $n(y - x) > 1$:

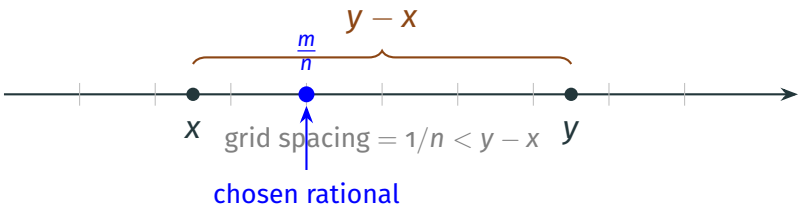
$$nx < m \leq 1 + nx < ny.$$

Since $n > 0$:

$$\boxed{x < \frac{m}{n} < y}$$

Set $p = m/n \in \mathbb{Q}$. $\square$

Proof of 1.20(b): Visual Summary



Existence of nth Roots

Theorem 1.21: Existence of nth Roots

Theorem 1.21

For every real $x > 0$ and every integer $n > 0$, there is one and only one positive real y such that

$$y^n = x.$$

This number y is written $\sqrt[n]{x}$ or $x^{1/n}$.

Proof Strategy: Theorem 1.21

Goal: Find $y > 0$ with $y^n = x$.

Approach:

1. **Uniqueness:** If $0 < y_1 < y_2$, then $y_1^n < y_2^n$. At most one root.
2. **Define** $E = \{t > 0 : t^n < x\}$.
3. **Show** $E \neq \emptyset$ and E is bounded above. Set $y = \sup E$.
4. **Rule out** $y^n < x$: build $h > 0$ with $(y + h)^n < x$, so $y + h \in E$ — contradicts y being an upper bound.
5. **Rule out** $y^n > x$: build $k > 0$ with $(y - k)^n > x$, so $y - k$ is an upper bound — contradicts y being the least upper bound.

Proof of Theorem 1.21 — Step 1: Uniqueness

Step 1: At most one such y

If $0 < y_1 < y_2$, then

$$y_1^n < y_2^n.$$

So there is at most one positive real y with $y^n = x$.

Proof of Theorem 1.21 — Step 2: Define E

Step 2: Define the set E

Let

$$E = \{t \in \mathbb{R} : t > 0 \text{ and } t^n < x\}.$$

E **is not empty:** Take $t = \dfrac{x}{1+x}$. Then $0 < t < 1$, so $t^n \leq t < x$.

E **is bounded above:** If $t > 1 + x$, then $t^n \geq t > x$, so $t \notin E$.

Proof of Theorem 1.21 — Step 3: $y = \sup E$

Step 3: Apply Theorem 1.19

Since $E \neq \emptyset$ and E is bounded above, Theorem 1.19 gives:

$$y = \sup E.$$

Claim: $y^n = x$.

Proof of Theorem 1.21 — Key Inequality

Key algebraic identity

For $0 < a < b$:

$$b^n - a^n = (b-a)\left(b^{n-1} + b^{n-2}a + \cdots + a^{n-1}\right)$$

The second factor has n terms, each $\leq b^{n-1}$, so:

$$\boxed{b^n - a^n < (b-a)\, n\, b^{n-1}}$$

Proof of Theorem 1.21 — Case: $y^n < x$

Case 1: Suppose $y^n < x$ — derive contradiction

Choose h with $0 < h < 1$ and

$$h < \frac{x - y^n}{n(y+1)^{n-1}}.$$

Set $a = y$, $b = y + h$. Then:

$$(y+h)^n - y^n < h\,n\,(y+h)^{n-1} < h\,n\,(y+1)^{n-1} < x - y^n.$$

So $(y+h)^n < x$, meaning $y + h \in E$.

But $y + h > y = \sup E$. Contradiction!

Proof of Theorem 1.21 — Case: $y^n > x$

Case 2: Suppose $y^n > x$ — derive contradiction

Put

$$k = \frac{y^n - x}{n\, y^{n-1}}.$$

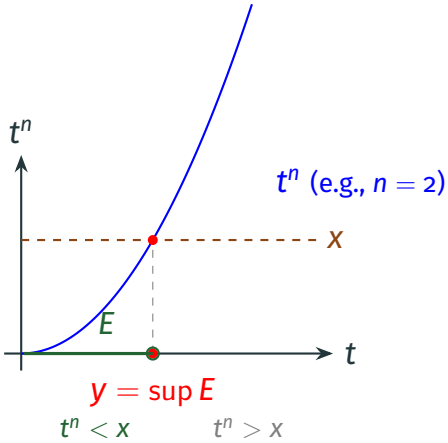
Then $0 < k < y$. By the key inequality (with $a = y - k$, $b = y$):

$$y^n - (y-k)^n < k\, n\, y^{n-1} = y^n - x,$$

so $(y-k)^n > x$. Hence $y - k$ is an upper bound of E.

But $y - k < y = \sup E$. Contradiction!

Proof of Theorem 1.21: Visual Summary



Corollary and Decimal Expansions

Corollary to Theorem 1.21

Corollary

If $a, b > 0$ are real and n is a positive integer, then

$$\boxed{(ab)^{1/n} = a^{1/n}\, b^{1/n}.}$$

Proof of the Corollary

Proof

Put $\alpha = a^{1/n}$ and $\beta = b^{1/n}$. Then:

$$ab = \alpha^n \beta^n = (\alpha\beta)^n$$

by commutativity of multiplication (Axiom M2, Definition 1.12).

By the uniqueness in Theorem 1.21:

$$(ab)^{1/n} = \alpha\beta = a^{1/n}\, b^{1/n}. \qquad \square$$

1.22: Decimals

Let $x > 0$ be real. Define a sequence of digits:

- n_0 = largest integer $\leq x$ (exists by archimedean property)
- n_k = largest integer such that

$$n_0 + \frac{n_1}{10} + \cdots + \frac{n_k}{10^k} \leq x \tag{5}$$

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Let E be the set of all such partial sums. Then $x = \sup E$.

The decimal expansion of x is:

$$n_0 \cdot n_1 n_2 n_3 \cdots \tag{6}$$

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- **Theorem 1.19:** $\mathbb{R}$ exists as an ordered field with the least-upper-bound property, containing $\mathbb{Q}$
- **Theorem 1.20(a):** Archimedean property — no infinitely large elements
- **Theorem 1.20(b):** $\mathbb{Q}$ is dense in $\mathbb{R}$ — rationals are everywhere

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- **Theorem 1.19:** $\mathbb{R}$ exists as an ordered field with the least-upper-bound property, containing $\mathbb{Q}$
- **Theorem 1.20(a):** Archimedean property — no infinitely large elements
- **Theorem 1.20(b):** $\mathbb{Q}$ is dense in $\mathbb{R}$ — rationals are everywhere
- **Theorem 1.21:** Every positive real has a unique nth root

Coming up next: The extended real number system ($+\infty$ and $-\infty$).